Naming compounds (1)

Most chemical **formulas** for **compounds** have numbers in them, as well as the symbols for different elements. These numbers tell us how many atoms of each element there are in one molecule of the compound.

Rule 1

When two elements are present:

- the metal is named first
- the elements will combine to made a compound ending in 'ide'
- if both are non-metals and one of them is hydrogen, hydrogen is written first.

For example:

KBr	potassium brom**ide**
MgO	magnesium ox**ide**
NaCl	sodium chlor**ide**
CaS	calcium sulf**ide**

Rule 2

When two or more elements combine with oxygen:

- the metal is named first
- the compound often ends with part of the name of the second element, combined with 'ate'.

For example:

Na_2CO_3	sodium carbon**ate**
$ZnSO_4$	zinc sulf**ate**
$NaNO_3$	sodium nitr**ate**
$MgSO_4$	magnesium sulf**ate**

Rule 3

When there are only two elements and both are non-metals:

- if hydrogen is present, it is written first
- prefixes are joined to the name of the second element to show how many atoms of it there are. The prefixes are:
  - di – 2 • tri – 3 • tetra – 4 • penta – 5.

  Mono means 1. The '1' is commonly left out, because scientists assume that no prefix means that there is only one of that atom.)

  For example:

CO_2	carbon **di**oxide	CO	carbon **mon**oxide
SO_3	sulfur **tri**oxide	HBr	hydrogen bromide

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Naming compounds (2)

Use the information on the Resource sheet 'Naming compounds (1)' to answer the following questions.

☞ **1** Fill in the missing words in the sentences below.

a When two e_____ combine, the name usually ends in 'i_____'.

b When two elements combine, the m_____ is written first.

c When two or more elements combine with o_____, the name of the compound usually ends in 'a_____'.

d If two elements are present and both are n_____-metals, we use a p_____ on the second element.

e 'Mono' means o_____.

2 Name the compound that magnesium will make with each of the following elements:

a chlorine _____

b oxygen _____

c fluorine _____

3 Complete the table below.

Rule number	Chemical formula	Name of compound
Rule 1	NaBr	
	KI	
	Na_2O	
Rule 2	Na_2CO_3	
	H_2SO_4	
	$CuSO_4$	
Rule 3	SO_2	
	CO	
	NO_3	

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